

Performance evaluation and optimization of a suspension-type reactor for use in heterogeneous catalytic ozonation

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ABSTRACT

Packed fixed-bed reactors are traditionally used for heterogeneous catalytic ozonation. However, a high solid-to-liquid requirement, poor ozone dissolution, ineffective utilization of catalyst surface area, and production of large amounts of catalyst waste impede application of such reactors. In this study, we designed a suspension catalytic ozonation reactor and compared the performance of this reactor with that of a traditional fixed-bed catalytic ozonation reactor employing oxalic acid (OA) as the target contaminant. Our results showed that total O_3 dissolved into the suspension reactor ($117\text{--}134\text{ mg.L}^{-1}$) was much higher compared to that measured in the fixed-bed reactor (53 mg.L^{-1}) due to a higher $O_3(g)$ interphase mass transfer rate in the suspension reactor. In accordance with the higher $O_3(g)$ interphase mass transfer, we observed a much higher proportional OA removal (32 %) compared to that achieved in the fixed-bed reactor (10%) employing an Fe-oxide catalyst supported on Al_2O_3 (Fe-oxide@ Al_2O_3) in both reactors. Use of a double-layered Cu-Al hydroxide (Cu-Al LDHs) catalyst in the suspension reactor further enhanced the performance with nearly 90 % OA removal observed. Given the superior performance of the suspension reactor, we investigated the impact of operating conditions (catalyst dosage, hydraulic retention time and ozone dosage) employing Cu-Al LDHs as the catalyst. We also developed a mathematical kinetic model to describe the performance of the suspension reactor and, through use of the kinetic model, showed that $O_3(g)$ interphase transfer rate was the rate-limiting step in OA removal. Thus, improvement in ozone gas diffuser design is required to improve the performance of the suspension reactor. Overall, the present study demonstrated that suspension reactors were more effective than fixed-bed reactors for oxidation of surface-active organic compounds such as OA due to the higher ozone interphase mass transfer rate and effective utilization of the catalyst surface area that can be achieved. As such, further research on suspension reactor design and development of catalysts suitable for use in suspension reactors should facilitate large-scale application of catalytic ozonation processes by the wastewater treatment industry.

1. Introduction

In the modern world, rapidly increasing urbanization and industrialization has resulted in the generation of a wide variety of recalcitrant organic pollutants (Ahmad et al., 2020; Wen et al., 2017). Contamination of water resources by these pollutants has resulted in significant threats to the environment and human health in many developing countries (Kumar et al., 2021; Mukhopadhyay et al., 2022). Advanced oxidation processes (AOPs) are a promising suite of technologies for the degradation of recalcitrant organics present in aquatic systems (Mishra et al., 2023; Shi et al., 2023). Of the various AOPs available, heterogeneous catalytic ozonation (HCO), wherein conversion of ozone to more

powerful oxidants such as hydroxyl radicals ($\cdot OH$) occurs via interaction of ozone with the catalyst surface, is one of the most promising technologies for wastewater treatment (Chen et al., 2023a, 2023b; Xu et al., 2023; Yu et al., 2020) due to its effectiveness in degrading refractory pollutants and enhancing the biodegradability of secondary effluent. As such, many researchers have investigated the use of HCO for treatment of recalcitrant organics over the past few decades. However, HCO is still not sufficiently developed to meet the needs of actual wastewater treatment on an industrial scale.

While a significant amount of work has been undertaken on the development of low-cost and effective catalysts for use in industrial scale HCO reactors research on both reactor design and overall process

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optimization has been limited. Packed fixed-bed reactors are traditionally used for HCO due to their ease of assembly and simple operation (Cao et al., 2021). However, a high solid-to-liquid requirement, poor ozone dissolution, ineffective utilization of catalyst surface area, and production of large amounts of catalyst waste impede application of such reactors (Tisa et al., 2014; Wang and Chen, 2020) on an industrial scale. In comparison, fluidized-bed or suspension catalytic ozonation reactors have been suggested to be more effective than fixed-bed reactors due to the increased accessibility of catalyst surface sites and the potential to operate at a relatively low solid-to-liquid ratio (Wei et al., 2020). For HCO, suspension reactors appear to be particularly promising since the gaseous ozone (which contains 85–90 % v/v oxygen (von Sonntag and von Gunten, 2012)) that is typically injected at the base of the system can assist in maintaining catalyst particles in suspension without the need for external aeration (as required in the fluidized bed reactor). Incorporation of an effective ozone injection system, a catalyst of suitable buoyancy and an ozone-resistant membrane system to retain the catalysts within the reaction chamber enables development of full scale treatment systems that are capable of continuous operation (Dai et al., 2021; Nandy et al., 2021).

While earlier studies have reported that suspension reactors are more effective than fixed-bed reactors (Tisa et al., 2014; Wei et al., 2020), a clear comparison of the performance of the two reactor types with respect to ozone mass transfer efficiency, ozone utilization efficiency and organic degradation efficiency is lacking. Thus, in this study we compare the performance of pilot scale fixed-bed and suspension HCO reactors for degradation of oxalic acid (OA). We used OA as the target contaminant since it is ozone resistant ($k_{O_3} = 0.04 \text{ M}^{-1} \cdot \text{s}^{-1}$) (Garoma and Gurul, 2005) and hence is not removed by the pure ozonation process. Furthermore, OA is directly oxidized to CO_2 and H_2O and does not form any intermediate products that would otherwise also scavenge $\cdot\text{OH}/\text{O}_3$ and complicate quantification of the efficacy of the process. For HCO in the fixed-bed reactor, we used Fe-oxide@ Al_2O_3 developed in our recent work as the catalyst (Kong et al., 2023) since this catalyst displayed high ($\sim 100\%$) OA removal efficiency in the batch system (Kong et al., 2023). For HCO in the suspension reactor, we used powdered Fe-oxide@ Al_2O_3 , prepared by grinding the catalyst used in fixed-bed reactor. Comparison of the performance of the same catalyst in two different reactors provides insight into the influence of the reactor design and gas-liquid hydrodynamics within the reactor on the performance of HCO. Note that the catalytic activity of the powdered and solid Fe-oxide@ Al_2O_3 composite is similar since the decay of ozone and associated removal of OA mainly occur via interaction with Fe-oxide and not Al_2O_3 since Al_2O_3 is inert towards O_3 and shows low sorption capacity for OA (Kong et al., 2023). We also used Cu-Al layered double hydroxides (Cu-Al LDHs) as the catalyst in the suspension reactor since Cu-Al LDHs was also effective in oxidation of OA during HCO in our recent study (Yuan et al., 2021). Cu-Al LDHs is an ideal candidate for use in a suspension reactor due to its high catalytic activity, low futile consumption of ozone, small catalyst size and low density with the latter two factors facilitating easy maintenance of particles in suspension. Based on the results obtained in the present study, we have compared the performance of the suspension reactor with that of a fixed-bed reactor with respect to catalyst usage, ozone usage and degradation efficiency. Furthermore, we have developed a mathematical model to describe the performance of the suspension reactor used here. The mathematical model developed can be used to predict the performance of the HCO process in a suspension reactor employing Cu-Al LDHs as the catalyst over a range of operating conditions. More importantly, the developed kinetic model can be used to optimize operational conditions of the suspension reactor such that degradation performance and ozone utilization efficiency are maximized.

2. Materials and methods

2.1. Chemicals and reagents

All chemicals, including copper(II) nitrate trihydrate ($\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$), sodium carbonate (Na_2CO_3), sodium bicarbonate (NaHCO_3), sodium hydroxide (NaOH) and nitric acid (HNO_3), were purchased from Sinopharm Chemical Reagents Co., Ltd. Aluminum nitrate nonahydrate ($\text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$) was purchased from Macklin Reagent Co., Ltd. All solutions were prepared in distilled water (OriginWater water purifier; D8008, Juyuanxing Technology Development, Beijing Co., Ltd.). The indigo stock solution was prepared by dissolving $0.6 \text{ g} \cdot \text{L}^{-1}$ potassium indigo trisulfonate in 20.0 mM phosphate buffer solution and stored in the dark prior to use. The ozone stock solution for batch experiments was prepared by continuously bubbling $\text{O}_3(\text{g})$ ($\sim 4\% \text{ O}_3(\text{g})/\text{O}_2(\text{g})$ mixture) for 30 min into distilled water (1 L). The ozone concentration in the stock solution was determined by UV spectroscopy at 253 nm (molar absorptivity = $3200 \text{ M}^{-1} \text{ cm}^{-1}$) (Bader and Hoigné, 1981).

All experiments were performed at a constant pH of 8.5 ± 0.1 employing 2 mM NaHCO_3 as the buffer. The pH of 8.5 is typical of many wastewaters and was the equilibrium pH of the 2 mM $\text{NaHCO}_3/0.03\%$ CO_2 buffer system employed in all studies described here. We used a bicarbonate buffer to control the pH since all real wastewaters would be expected to contain bicarbonate ions. Even though bicarbonate ions are known scavengers of $\cdot\text{OH}$ (Vione et al., 2009), the scavenging of $\cdot\text{OH}$ by bicarbonate ions during HCO was not important since the catalyst- O_3 interaction (for both Cu-Al LDHs and Fe-oxide@ Al_2O_3 catalyst) mainly results in the formation of surface-associated $\cdot\text{OH}$ (Kong et al., 2023; Yuan et al., 2021). As such, the scavenging of $\cdot\text{OH}$ by solution phase species such as bicarbonate ions was minimal. Solutions of 2 mM NaOH and 2 mM HNO_3 were prepared for initial pH adjustment if required. Synthetic wastewater containing 2 mM NaHCO_3 and $1\text{--}2 \text{ mM}$ of OA was prepared for testing the performance of the fixed-bed and suspension reactors.

2.2. Cu-Al LDHs synthesis

A similar approach to the synthesis of Cu-Al LDH as described in our earlier work (Yuan et al., 2021) was used here with the only modification being a reduction in the aging time of the Cu-Al LDHs from 24 h to 3 h to enhance the rate of production of catalyst. Briefly, a solution containing 100 mM of both $\text{Cu}(\text{NO}_3)_2$ and $\text{Al}(\text{NO}_3)_3$ was added dropwise into a solution containing 200 mM Na_2CO_3 under continuous stirring at controlled pH of 10 (achieved by addition of 1 mM NaOH). The obtained gel was placed in an oil bath (Shanghai XIUI lab instrument Co., Ltd) at 60°C for 3 h followed by washing with distilled water several times. The washed slurry was dried overnight at 40°C and ground using a mortar and pestle to generate fine particles of Cu-Al LDHs.

2.3. Fe-oxide synthesis

As previously described, a urea-based heterogeneous precipitation method was used for preparation of the Fe-oxide@ Al_2O_3 composite catalyst (Kong et al., 2023). Briefly, Al_2O_3 pellets were calcined in a muffle furnace at 450°C for 3 h and then immersed in a metal-precursor solution containing 10 mM $\text{Fe}(\text{NO}_3)_3$ and 15 mM urea (pH 2.3), followed by placing in a vacuum environment for 48 h. The impregnated catalyst particles produced following supernatant decantation were aged for 12 h before being calcined for 3 h at 450°C . The obtained catalyst was stored in the dark prior to use. Note that the catalyst prepared in our earlier study (Kong et al., 2023) was not calcinated while the catalyst prepared here was calcinated in order to improve its stability over long-term use. For use in the suspension reactor, the prepared Fe-oxide@ Al_2O_3 composite catalyst was crushed into a fine powder using a mortar and pestle.

2.4. Cu-Al LDHs and Fe-oxide@Al₂O₃ characterization

The morphological characterization of the synthesized Cu-Al LDHs and Fe-oxide@Al₂O₃ catalysts was performed using scanning electron microscopy-energy dispersive spectroscopy (SEM-EDS). The chemical characterization of the catalyst was performed employing X-ray powder diffraction (XRD) and Fourier transform infrared spectroscopy (FTIR, Perkin Elmer Frontier IR spectrometer). The surficial area was determined by analysis of N₂ sorption isotherm data using the Brunauer–Emmett–Teller (BET) model. The size distribution and average diameter of the Cu-Al LDHs particles were obtained using a Micromeritics NanoPlus HD-3 instrument. The viscosity of the Cu-Al LDHs suspension was measured using a Brookfield Ultra Rheometer with shear rates ranging from 0 to 300 s⁻¹ at a controlled temperature of 25 °C. The surface hydroxyl group density of both catalysts was measured using a saturated-protonation method (Davranche et al., 2003; Tamura et al., 1999; Zhang et al., 2008). Specifically, the catalyst (0.3 g.L⁻¹ of Cu-Al LDHs or 500 g.L⁻¹ of Fe-oxide@Al₂O₃) was added to solutions containing varying concentrations of NaOH (0.01–0.1 M) and allowed to sit for 24 h. Subsequently, the suspensions were filtered using PVDF membrane filters (0.22 µm) and the residual NaOH concentration in the filtrates measured by titration using 0.1 M HNO₃ solution. The concentration of the acidic surface hydroxyl groups was the same as the concentration of NaOH consumed during this acid-base titration and total hydroxyl group concentration was twice of the acidic hydroxyl group concentration based on charge balance. The stability of the catalyst in the presence of ozone was also measured by continuous bubbling of 60 mg.L⁻¹ O₃(g) at a flow rate of 3 L/h into the catalyst suspension for 3 h and the concentration of Cu and Fe leached was measured using inductively coupled plasma–optical emission spectrometry (ICP-OES).

2.5. Reactor design

Schematics of the suspension and fixed-bed reactors are shown in Fig. 1. The total volume of the suspension-type reactor was 40 L and

consisted of three-zones, namely, (i) a pre-mixing zone wherein catalyst and wastewater are mixed, (ii) a main reaction zone where ozone dissolution and subsequent organic degradation via HCO occurs, and (iii) a filtration zone in which clean permeate is produced by use of α-Al₂O₃ ceramic flat sheet membrane modules provided by Sicer Membrane Technology Co., Ltd, China. If need be, a portion of the catalyst can also be continuously replaced in this filtration zone. A small mixer (4 cm diameter) with rotor speed of 300 rpm was installed 8 cm above the base of the reactor in the pre-mixing zone in order to ensure adequate mixing of catalyst particles. Two different types of diffusers for injection of the O₃(g)/O₂(g) mixture were installed in the main reaction zone to supply ozone and to maintain the catalyst particles in suspension. An oval-shaped titanium-based diffuser of 5 cm diameter was employed to cover the middle area of the reactor while a bar diffuser (length: 16 cm) was used to minimize Cu-Al LDHs/Fe-oxide@Al₂O₃ settling near the walls of the reactor. Additionally, a mixer (7.5 cm diameter) with rotational speed of 5–10 rpm was employed in the main reaction zone (placed 3 cm above the bottom of the reactor) to assist further in maintaining the Cu-Al LDHs//Fe-oxide@Al₂O₃ particles in suspension, particularly in the low-liquid velocity regions near the walls of the reactor. Based on calculation of settling velocity, particle relaxation time and Froude's number, we determined that the mixing was adequate to maintain the catalyst particle in suspension (see Supporting Information Section S1 for details of the calculations).

The volume of the fixed-bed reactor was 2 L and consisted of two-zones, namely (i) a pre-mixing zone, and (ii) the main reaction zone with catalyst bed. The pre-mixing and main reaction zones were separated by a porous catalyst plate holder with the size of the two zones able to be varied by adjusting the position of the catalyst plate holder. A pore diameter of 0.3 cm was used for the catalyst holder to avoid blockage/disruption of the bubbles entering the catalytic zone from the pre-mixing zone. An ozone diffuser, with ceramic base (5 cm diameter) and mineral stone diffuser plate (7 cm diameter), was placed in the pre-mixing zone at the base of the reactor.

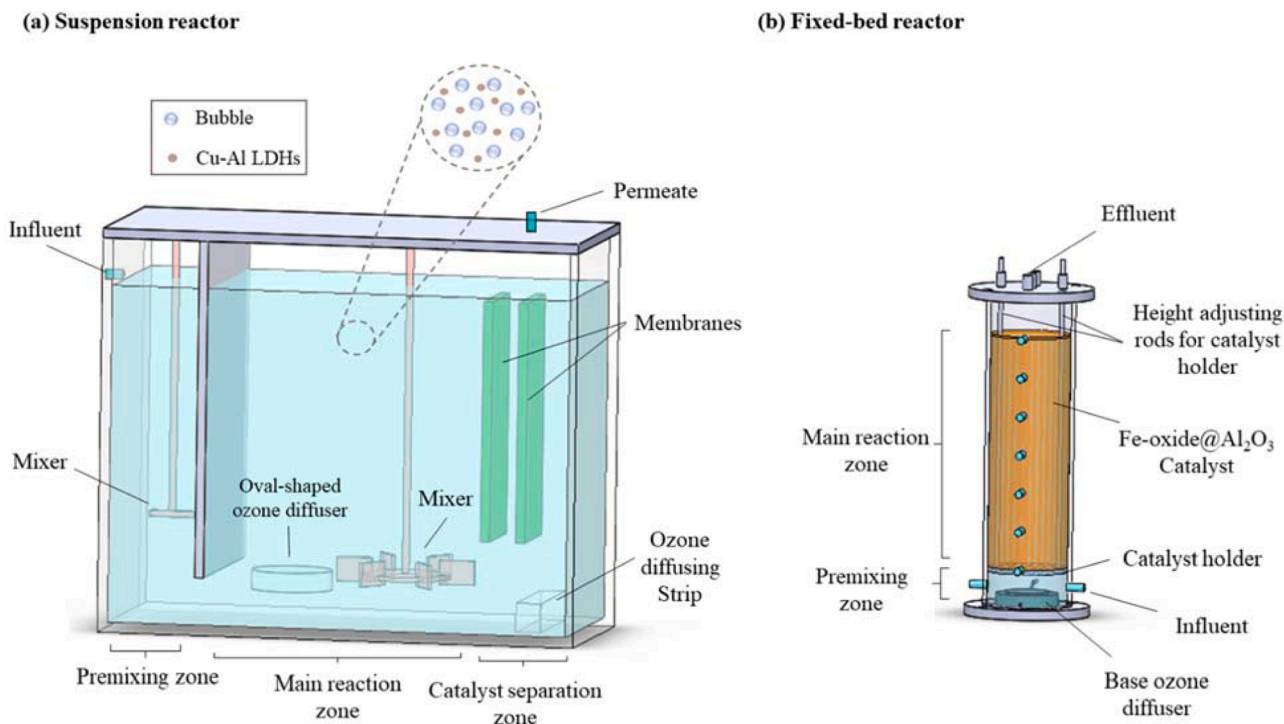


Fig. 1. Schematics of the suspension reactor with dimensions of 22 cm length, 40 cm height and 25 cm width (a) and the fixed-bed reactor with dimensions of 32 cm height and 9 cm inner diameter (b).

2.6. Measurement of ozone dissolution

To determine the ozone dissolution efficiency of the suspension and fixed-bed reactors, the extent of ozone dissolution in the reactors was quantified in the absence of OA. For measurement, the reactors (both suspension and fixed-bed) were filled with 2 mM NaHCO₃ solution and either 0.1 g.L⁻¹ Cu-Al LDHs (for suspension type) or 1 g.L⁻¹ powdered Fe-oxide@Al₂O₃ (for suspension type) or 1000 g.L⁻¹ Fe-oxide@Al₂O₃ (for fixed-bed type). Note that the Fe-oxide@Al₂O₃ composite contains ~10 % Fe by mass (Kong et al., 2023); hence a concentration of 1 g.L⁻¹ Fe-oxide@Al₂O₃ is comparable to a Cu-Al LDHs concentration of 0.1 g.L⁻¹ in terms of active metal concentration. Subsequently, O₃(g) at a concentration of 60 mg.L⁻¹ was continuously bubbled into the suspension reactor at an O₃(g) flow rate of 3 L/min and a feed flow rate of 40 L/h. For the fixed-bed reactor, O₃(g) at a concentration of 60 mg.L⁻¹ was continuously bubbled into the fixed-bed reactor at an O₃(g) flow rate of 0.15 L/min and a feed flow rate of 2 L/h. At desired time intervals, a 1 mL sample was taken from the effluent and dissolved O₃ concentration in the sample was measured using the indigo method (Bader and Hoigné, 1981). Detailed description of the dissolved O₃ concentration measurement using the indigo method is provided in SI Section S2. The outlet O₃(g) concentration was also measured using an O₃ gas analyser (PGD3-H-O₃, Shenzhen Xinhairui Technology Development Co., Ltd). The total transferred ozone dosage (TOD), which represents the accumulated amount of ozone transferred to solution per unit volume for both units, was calculated using Eq. (1). The pseudofirst-order decay rate constant (k_d) of dissolved O₃ in both units was determined using the mass balance equation (Eq. (2)).

$$\text{TOD} = \frac{F_g}{F_l} ([O_3(g)]_{in} - [O_3(g)]_{out}) \quad (1)$$

where F_g and F_l represent the O₃(g) flow rate and feed flow rate, respectively. $[O_3(g)]_{in}$ and $[O_3(g)]_{out}$ represent the inlet and outlet O₃(g) concentrations, respectively.

$$F_g ([O_3(g)]_{in} - [O_3(g)]_{out}) = k_d C_l V \quad (2)$$

where C_l and V represent the dissolved ozone concentration in the reactor and volume of the reactor, respectively. The remaining terms are the same as described above.

The interphase ozone mass transfer coefficient ($K_l a$), which controls the rate and extent of ozone transferred from the gas phase to the bulk liquid (Eq. (3)) (Lian et al., 2022) for both units, was also calculated using Eq. (4) (see SI section S3 for details of the derivation of this equation):

$$S_{O_3}^l = K_l a (C_l^* - C_l) \quad (3)$$

$$K_l a = \frac{C_l}{C_l^* - C_l} k_d \quad (4)$$

where $S_{O_3}^l$ represent the O₃(g) interphase mass transfer rate, $K_l a$ is the ozone interphase mass transfer coefficient and C_l^* is the saturation concentration of dissolved ozone in the reactor calculated employing Henry's law (Eq. (5)) and C_l is the measured dissolved ozone concentration.

$$C_l^* = K_H P_g \quad (5)$$

where K_H is Henry's law constant (0.01 mol kg⁻¹ bar⁻¹ at 20 °C) (Lian et al., 2022) and P_g is the partial pressure of O₃(g) in the reactor.

2.7. Measurement of OA oxidation

For measurement of OA oxidation in the suspension reactor, the reactor was filled with 40 L of synthetic wastewater (containing 2 mM NaHCO₃ and OA) and 0.1–1 g.L⁻¹ Cu-Al LDHs or 1 g.L⁻¹ Fe-

oxide@Al₂O₃. Subsequently, O₃(g) at a concentration of 20–60 mg.L⁻¹ was continuously bubbled into the reactor at an O₃(g) flow rate of 1–3 L/min. A feed flow rate of either 20 L/h or 40 L/h was used to yield hydraulic residence times (HRT) of 2 h or 1 h, respectively. At desired time intervals, 5 mL of sample was withdrawn from the main reaction zone as well as from the effluent and the samples were sparged with air for 30 s to remove any dissolved O₃ present. The samples collected from the reaction zone were filtered using PVDF membrane filters (0.22 µm) and the total organic carbon (TOC) content of the samples (both reaction zone and effluent) was measured using a TOC analyser (TOC-L CPH, SHIMADZU, Japan). The outlet O₃(g) concentration was measured using the O₃ gas analyser.

For measurement of OA oxidation in the fixed-bed reactor, the reactor was filled with catalyst (~1000 g.L⁻¹ of catalyst) and synthetic wastewater. Subsequently, O₃(g) at a concentration of 60 mg.L⁻¹ was continuously bubbled into the reactor at an O₃(g) flow rate of 0.15 L/min. A feed flow rate of 2 L/h was used to yield an HRT of 1 h. At desired times, 5 mL of effluent was collected and sparged with air for 30 s to remove any dissolved O₃ present. Subsequently, samples were filtered using 0.22 µm PVDF filters and the TOC content of the samples was measured. The outlet O₃(g) concentration was measured using the O₃ gas analyser. The performance of the pure ozonation process in both pilot units was also measured employing the same process conditions as described above except without the addition of catalyst.

All experiments in both units were continued until steady state was achieved with all experiments being performed in triplicate. No variation in the pH was observed during experiments. We would like to highlight that even though different O₃(g) flow rates and liquid flow rates were used for the suspension and fixed-bed reactors, the quantity of O₃(g) used per unit volume of feed, the O₃:DOC concentration ratio and the HRT were similar for both suspension and fixed-bed reactors, making the comparison of results from the two systems meaningful.

2.8. Measurement of O₃ decay and OA removal by Cu-Al LDHs and Fe-oxide@Al₂O₃ in batch reactor

To compare the performance of the Fe-oxide@Al₂O₃ and Cu-Al LDHs catalysts with respect to ozone decay and OA removal under the same reactor condition, measurement of O₃ decay and OA removal by HCO was performed in a batch reactor using the method described in our earlier works (Kong et al., 2023; Yuan et al., 2021). For measurement of O₃ decay, an appropriate volume of dissolved ozone stock solution was added to the reactor containing 2.0 mM NaHCO₃ solution and catalyst (0.1 g.L⁻¹ Cu-Al LDHs and 60 g.L⁻¹ Fe-oxide@Al₂O₃) to yield a final dissolved ozone concentration of 100 µM. At appropriate times, samples were withdrawn from the reactor, filtered using 0.22 µm PVDF filters, and then dissolved ozone concentration was measured using the indigo method (Bader and Hoigné, 1981).

For measurement of OA removal, an appropriate volume of OA stock solution was added to 70.0 mL of pH 8.5 buffer solution containing either 0.1 g.L⁻¹ Cu-Al LDHs or 60 g.L⁻¹ Fe-oxide@Al₂O₃, resulting in a final OA concentration of 1.0 µM (0.1 µM radiolabelled and 0.9 µM non-radiolabelled OA). Subsequently, 10.0 µM dissolved ozone was added to the reactor to initiate the reaction. At regular intervals, 1.5 mL of sample was extracted from the reactor and the concentration of residual radiolabelled OA in the sample was measured using the liquid scintillation counting method. See SI section S2 for detailed description of the method.

2.9. Kinetic modelling

Kinetic modelling of the performance of the suspension reactor was performed employing the software package Kintecus (Ianni, 2020). The model for OA oxidation by the Cu-Al LDHs catalyst in the batch system was developed in our earlier study (Yuan et al., 2021) and is shown in Table 1. Please refer to our earlier work (Yuan et al., 2021) for a detailed

Table 1

Developed kinetic model for ozone decay and OA oxidation.

No	Reaction	Rate constant ($M^{-1}.s^{-1}$)	Published value ($M^{-1}.s^{-1}$)
O ₃ decay in solution			
1	$O_3 + OH^- \rightarrow HO_2 + O_2^-$	1.2×10^2	70–100
2	$O_3 + O_2^- \rightarrow HO_3 + O_2$	1.5×10^9	1.5×10^9
3	$HO_3 / O_3^- \rightarrow OH + O_2$	4.8×10^4	4.8×10^4
4	$OH + O_3 \rightarrow O_2^- + O_2$	1.0×10^8	1.0×10^8 – 2.0×10^9
5	$O_3 + CO_3^{2-} \rightarrow HCO_3^- + O_2$	1.0×10^5	1.0×10^5
6	$OH + H_2O_2/HO_2^- \rightarrow H_2O + O_2^-$	3.4×10^7	2.7×10^7
7	$OH + H_2CO_3/HCO_3^-/CO_3^{2-} \rightarrow H_2O + CO_3^{2-}$	1.4×10^7	8.2×10^6
8	$CO_3^{2-} + H_2O_2/HO_2^- \rightarrow O_2^- + HCO_3^-$	4.6×10^5	4.3×10^5
9	$CO_3^{2-} + CO_3^{2-} \rightarrow CO_4^{2-} + HCO_3^-$	2.0×10^7	2.0×10^7
OA oxidation in solution			
10	$C_2O_4^{2-} + OH^- \rightarrow C_2O_4^- + H_2O$	7.7×10^6	7.7×10^6
11	$C_2O_4^{2-} + CO_3^{2-} \rightarrow C_2O_4^- + HCO_3^-$	3.0×10^4	3.0×10^4
12	$C_2O_4^- \rightarrow CO_2^- + HCO_3^-$	1.0×10^9	Fast
13	$CO_2^- + O_2 \rightarrow O_2^- + HCO_3^-$	4.2×10^9	4.2×10^9
O ₃ decay via interaction with Cu-Al LDHs			
14	$O_3 + \equiv \rightleftharpoons \equiv O_3$	$K = 4.1 \times 10^{3a,b}$	$4.1 \times 10^{3a,b}$
15	$\equiv O_3 \rightarrow \equiv OH$	5.0×10^{-3}	5.0×10^{-3}
16	$\equiv O_3 + \equiv \rightarrow NRP + \equiv^c$	3.0	3.0
17	$\equiv OH + \equiv \rightarrow NRP + \equiv^c$	1.0×10^6	1.0×10^6
OA oxidation on Cu-Al LDHs surface			
18	$C_2O_4^{2-} + \equiv \rightleftharpoons \equiv C_2O_4^-$	$K = 2.4 \times 10^{4a,b}$	
19	$\equiv C_2O_4^{2-} + \equiv O_3 \rightarrow \equiv C_2O_4^- + HO_3^-$	1.0×10^8	1.0×10^6
20	$\equiv C_2O_4^{2-} + \equiv OH^- \rightarrow \equiv C_2O_4^- + \equiv$	1.0×10^9	1.0×10^9
21	$\equiv C_2O_4^- \rightarrow 2 CO_2 + \equiv$	1.0×10^9	1.0×10^9

^a based on surface site concentration of 36.7 $\mu\text{moles.g}^{-1}$ in Cu-Al LDHs.^b K represents the ratio of the forward adsorption rate constant and back desorption rate constant and has units of M^{-1} .^c NRP represents non reactive product.

description of these reactions. A brief description of these reactions is also included in SI Section S4. Herein, we combine the kinetic model shown in Table 1 with ozone dissolution within the reactor to predict OA removal in the suspension reactor. The amount of ozone transferred from the gas phase to the bulk liquid was simulated using Eq. (3). The mass transfer equation was simulated using the user-defined function available in Kintecus. Note that the saturated ozone concentration was calculated based on the ozone gas partial pressure in the reactor (and not the inlet ozone gas partial pressure). The value of $K_1\alpha$ (which depends on gas volume fraction, bubble size and ozone diffusivity) for the reactor unit used here was deduced from the measured dissolved O₃ concentration and O₃(g) concentration in the suspension reactor (Fig. 3) employing Eq. (4).

3. Results and discussion

3.1. Characterization of Cu-Al LDHs and Fe-oxide@Al₂O₃

As can be seen from the SEM image in Fig. 2a, the Fe-oxide@Al₂O₃ particles were of pseudo-spherical shape. In comparison, the Cu-Al LDHs particles exhibited a petal-like morphology and, based on the results of elemental mapping, exhibited a uniform distribution of Cu, Al and O (Fig. 2c–f) in agreement with previously published studies (Cha et al., 2020; Ye et al., 2020). The XRD spectra of both catalysts (Figs. 2b and 2g) confirmed that the main mineral phases of the Fe-oxide@Al₂O₃ and Cu-Al LDHs catalysts were $\alpha\text{-Fe}_2\text{O}_3 + \text{Al}_2\text{O}_3$ (JCPD#44–1415 and 10–0425) and Cu-Al carbonate hydroxide hydrate (JCPD#46–0099), respectively. Note that the Fe-oxide catalyst synthesized in our earlier work (Kong et al., 2023) consisted of $\gamma\text{-FeOOH}$ and $\alpha\text{-Fe}_2\text{O}_3$ on the Al₂O₃ support; however, only $\alpha\text{-Fe}_2\text{O}_3$ was shown to be present here, presumably due to the transformation of $\gamma\text{-FeOOH}$ to $\alpha\text{-Fe}_2\text{O}_3$ during

calcination (which was not performed in our earlier work). Even though $\alpha\text{-Fe}_2\text{O}_3$ was shown to be less effective than catalysts comprised of both $\gamma\text{-FeOOH}$ and $\alpha\text{-Fe}_2\text{O}_3$ (Kong et al., 2023), the use of $\alpha\text{-Fe}_2\text{O}_3$ @Al₂O₃ is acceptable since this catalyst also shows catalytic activity with much higher OA removal compared to that achieved in the pure ozonation process in the batch reactor (Fig. S1).

As shown in Fig. S2, the surface areas of the Fe-oxide@Al₂O₃ and Cu-Al LDHs catalysts were determined to be 206.7 m^2g^{-1} and 41.8 m^2g^{-1} , respectively. In addition, the diameters of the Fe-oxide@Al₂O₃ and Cu-Al LDHs particles were determined to be 3–5 μm and $5.9 \pm 0.2 \mu\text{m}$, respectively. The surface hydroxyl group densities on Fe-oxide@Al₂O₃ and Cu-Al LDHs were 155.6 and 36.7 $\mu\text{moles.g}^{-1}$, respectively. The higher surface hydroxyl group density of Fe-oxide@Al₂O₃ compared to Cu-Al LDHs can be attributed, at least in part, to the higher surface area of Fe-oxide@Al₂O₃. Importantly, continuous bubbling of O₃(g) for 3 h in suspensions containing either 500 g.L^{-1} of Fe-oxide@Al₂O₃ or 1 g.L^{-1} Cu-Al LDHs at pH 8.5 resulted in no leaching of Cu or Fe into the solution with this result supporting the conclusion that the catalysts prepared are quite stable.

3.2. Comparison of catalytic activity of Cu-Al LDHs and Fe-oxide@Al₂O₃ in facilitating O₃ decay and OA removal in batch reactor

Prior to investigating the performance of the fixed-bed and suspension reactors, we measured the performance of the Cu-Al LDHs and Fe-oxide@Al₂O₃ catalysts under the same conditions using a batch reactor to justify the use of two different catalysts in the suspension and fixed-bed reactors. As shown in Fig. S1a, Fe-oxide@Al₂O₃ is more effective in facilitating ozone decay compared to Cu-Al LDHs in the batch reactor, at least at the concentration investigated here. In comparison, a lower rate of OA removal is observed in the presence of Fe-oxide@Al₂O₃ (initial pseudo first-order rate constant = $0.023 \pm 0.005 \text{ min}^{-1}$ for 60 g.L^{-1}) compared to Cu-Al LDHs ($0.158 \pm 0.001 \text{ min}^{-1}$ for 0.1 g.L^{-1}) (Fig. S1b). However, in the fixed-bed reactor, a much higher dosage of Fe-oxide@Al₂O₃ (1000 g.L^{-1}) is used and thus the OA removal kinetics in the fixed-bed reactor (0.38 min^{-1}) is calculated to lie within the range of OA removal kinetics in the suspension reactor (0.158 – 1.58 min^{-1} for a Cu-Al LDHs loading of 0.1–1.0 g.L^{-1}) based on extrapolation of the measured kinetics of OA removal in the batch reactor to the catalyst dosage used in the fixed-bed and suspension reactors. On this basis, we consider it is reasonable to compare the performance of the Cu-Al LDHs catalyst in the suspension reactor with that of the Fe-oxide@Al₂O₃ catalyst used in the fixed-bed reactor.

3.3. Comparison of ozone mass transfer and OA removal in suspension and fixed-bed reactors

As shown in Figs. 3a & b, a higher TOD value and higher dissolved O₃ concentration are achieved in the suspension reactor (for both Fe-oxide@Al₂O₃ and Cu-Al LDHs catalyst) compared to that in the fixed-bed reactor in the presence of catalyst. Note that the TOD and dissolved O₃ concentration were similar in both the reactors in the absence of catalyst. The higher TOD value in the suspension reactor compared to that in the fixed-bed reactor in the presence of catalyst is due to a higher O₃(g) interphase mass transfer rate with $K_1\alpha$ values of 3.6 – $6.1 \times 10^{-3} \text{ s}^{-1}$ and $9.1 \times 10^{-4} \text{ s}^{-1}$ calculated for the suspension and fixed-bed reactors, respectively. A higher TOD in the suspension reactor compared to that measured in the fixed-bed reactor when Fe-oxide@Al₂O₃ is used as the catalyst in both reactors suggests that the lower O₃(g) interphase mass transfer rate in the fixed-bed reactor is not due to the difference in the nature of the catalyst but, instead, is caused by the difference in the gas-liquid hydrodynamics within the two reactors. Furthermore, the similarity in TOD and dissolved ozone concentrations in the two reactors in the absence of catalyst suggests that the difference in the gas-liquid hydrodynamics within the fixed-bed and suspension reactors is caused by the presence of the catalyst. The presence of the packed catalytic bed

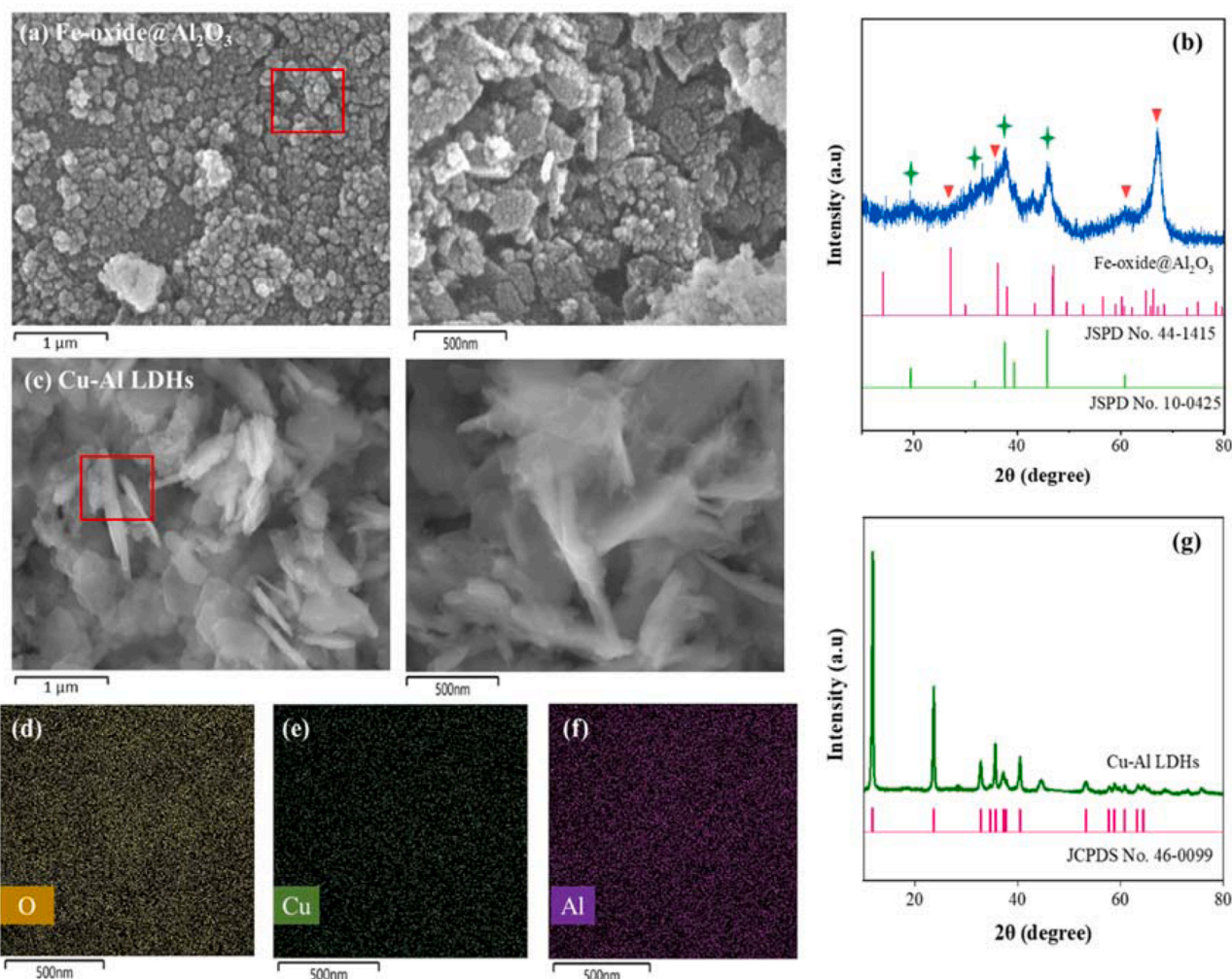


Fig. 2. Physiochemical characterization of synthesized catalysts. (a) SEM image of Fe-oxide@Al₂O₃, (b) XRD spectrum of Fe-oxide@Al₂O₃, (c) SEM and element mapping images of Cu-Al LDHs (d-f), and (g) XRD spectrum of Cu-Al LDHs.

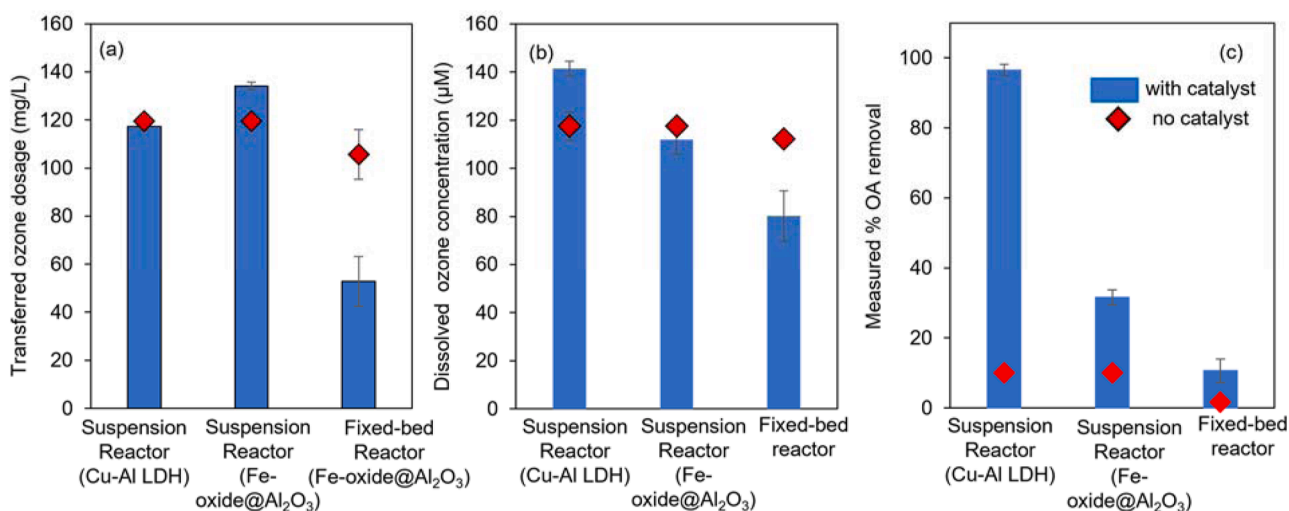


Fig. 3. Measured transferred ozone dosage (a), dissolved ozone concentration (b) and OA removal% (c) in the fixed-bed and suspension reactors in the absence (red diamonds) and presence (blue bars) of catalyst. Experimental conditions: O₃(g) = 60 mg.L⁻¹; [OA] = 1 mM; O₃: DOC ~ 12 mg/mg; HRT = 1 h, catalyst dosage = 0.1 g.L⁻¹ (for Cu-Al LDHs), 1 g.L⁻¹ (for Fe-oxide@Al₂O₃ in suspension reactor) and 1000 g.L⁻¹ (for Fe-oxide@Al₂O₃ in fixed-bed reactor).

causes bubble retention (and subsequent coalescence) with resultant low rate of $O_3(g)$ interphase mass transfer throughout the bed (Grace and Sun, 1991; Werther, 1984). The dissolved O_3 decay rate constants in the presence of Cu-Al LDHs in the suspension reactor and Fe-oxide@ Al_2O_3 in the fixed-bed reactors are comparable ($4.8 \pm 0.3 \times 10^{-3} s^{-1}$ and $3.8 \pm 0.4 \times 10^{-3} s^{-1}$, respectively) and are close to the value determined in the absence of catalyst ($5.2 \pm 0.3 \times 10^{-3} s^{-1}$). In comparison, a higher dissolved O_3 decay rate constant ($6.9 \pm 0.2 \times 10^{-3} s^{-1}$) in the presence of powdered Fe-oxide@ Al_2O_3 was observed in the suspension reactor. The lack of influence of Cu-Al LDHs on the rate of ozone decay agrees with our earlier work (Yuan et al., 2021) and the data presented in Fig. S1a which shows minimal (10 %) influence of Cu-Al LDHs on the ozone decay kinetics. However, the rate of O_3 decay kinetics in the presence of Fe-oxide@ Al_2O_3 is expected to be higher than that observed in the pure ozonation system based on the data shown in Fig. S1a as well as the measured O_3 decay kinetics in the suspension reactor containing Fe-oxide@ Al_2O_3 . Thus, it appears that the nature of the packed-bed structure results in ineffective utilization of the catalytic surface area with minimal ozone-catalyst interaction and associated O_3 decay.

In accordance with the higher rate of $O_3(g)$ interphase mass transfer, we observe a much higher OA removal in the suspension reactor compared to that in the fixed-bed reactor (Fig. 3c). As shown, nearly 32% and 100% of OA is removed in the suspension reactor employing Fe-oxide@ Al_2O_3 and Cu-Al LDHs as the catalyst while ~10% of the OA is removed in the fixed-bed reactor. Note that <10% OA removal by pure ozonation was observed in both the reactors confirming that OA removal occurs mainly via HCO. The improved $O_3(g)$ interphase mass transfer efficiency and increased access to active sites in the suspension reactor compared to the fixed-bed reactor enhances OA removal. The ozone utilization efficiency (i.e. g DOC removed per g of ozone consumed) was also higher in the suspension reactor compared to the fixed-bed reactor. A much higher DOC removal per g of catalyst used (24.0 mg DOC. g^{-1} for Cu-Al LDHs and 7.2 mg DOC. g^{-1} for Fe-oxide@ Al_2O_3) was observed in the suspension reactor compared to that in the fixed-bed reactor (9.6 μg DOC. g^{-1} catalyst). Even when normalized to the active site concentration (i.e. surface hydroxyl group concentration) rather than the bulk catalyst mass, the performance of the suspension reactor (0.65 mg DOC removed per $\mu mole$ Cu-Al LDHs active sites and 0.047 mg DOC removed per $\mu mole$ Fe-oxide@ Al_2O_3 active sites) was much higher than that of the fixed-bed reactor (0.06 μg DOC removed per $\mu mole$ Fe-oxide@ Al_2O_3 active sites). The significant difference in Fe-oxide@ Al_2O_3 performance in the suspension reactor and fixed-bed reactor with respect to OA removal further confirms that the packed-bed structure results in ineffective utilization of the catalyst surface area.

The combined catalyst and energy cost of the fixed-bed reactor under the conditions used here was calculated to be 0.11 \$/L, much higher than that of the suspension reactor (0.0085 \$/L), with the higher cost of the fixed-bed reactor due particularly to the large amount of catalyst used in the fixed-bed reactor (details of the cost calculation are provided in SI Section S5). Given the superior performance of the suspension reactor compared to the fixed-bed reactor, we investigate the impact of operating conditions on the performance of the suspension reactor in more detail in the following section. Note that since Cu-Al LDHs showed better performance than Fe-oxide@ Al_2O_3 catalyst in suspension reactor, we used Cu-Al LDHs as the catalyst in the following experiments.

3.4. Flow characteristics within the suspension reactor

We characterized the flow behavior inside the suspension reactor using Cu-Al LDHs as the catalyst since maintenance of the catalyst in suspension and good mixing within the reactor are critical to the performance of the reactor. The calculated settling velocity ($0.004 ms^{-1}$; see SI Section S1 for detailed calculation) of Cu-Al LDHs in the reactor was lower than the upward bubble velocity ($0.01 ms^{-1}$), suggesting that settling of Cu-Al LDHs should be minimal under the conditions used

here. A short relaxation time, which characterizes the particle response to changes in surrounding flow conditions (Vélez-Cordero et al., 2012), of 8.5 s further suggests that these particles readily adapt to flow changes and resist settling (Torres et al., 2015). Furthermore, the Froude number (i.e., the ratio of inertial forces to gravitational forces) was calculated to be 0.99 indicating that inertial forces were in balance with the gravitational forces, creating conditions for the particles to remain in suspension (Fox et al., 2020).

The rheological properties including viscosity, shear rate and shear stress also impact the maintenance of the particles in suspension (Khandavalli et al., 2019). As shown in Fig. S3a, the measured viscosity of the Cu-Al LDHs suspension was the same at different Cu-Al LDHs concentrations (0.1, 0.5, 1 and 2 g/L) with this result confirming that the presence of Cu-Al LDHs had no significant impact on the viscosity of the system. This observation further suggests that the loading of Cu-Al LDHs particles used does not significantly influence the rheology of the system (Guy et al., 2015). Increasing the shear rate from 0 to $300 s^{-1}$ enhanced the viscosity of the Cu-Al LDHs suspension, affirming the shear-thickening behavior of the particle-fluid mixture. The obtained shear-thickening behavior has implications to the stability of the particle suspension, particle settling behavior and homogeneity of the particle dispersion (Khandavalli et al., 2018). As presented in Fig. S3b, the increase in shear rate enhanced the shear stress experienced by the particles resulting in improved particle dispersion and diminished particle settling (Shi et al., 2020). Our shear stress and shear rate data, obtained at catalyst concentrations of 0.1 and $1.0 g.L^{-1}$, show good fits to the Herschel-Bulkley model (Burgos et al., 1999) (Figs. S3c and d) indicating that both suspensions behave as non-Newtonian fluids (Liu et al., 2015).

In summary, all observed rheological behavior strongly endorsed the use of Cu-Al LDHs in the suspension reactor.

3.5. Impact of various operating conditions on the performance of suspension reactor

As shown in Fig. 4a, at a Cu-Al LDHs concentration of $0.1 g.L^{-1}$ and feed OA concentration of 1 mM, OA removal increased with increase in the inlet O_3 :DOC concentration ratio achieved by either increasing the feed $O_3(g)$ concentration ($20-60 mg.L^{-1}$) or the $O_3(g)$ flow rate ($1-3 L/min$). These results confirm that ozone dosage (O_3 :DOC concentration ratio) is an important parameter with OA removal limited by O_3 availability. A similar trend is observed at a Cu-Al LDH dosage of $1.0 g.L^{-1}$ (Fig. 4b and 4c; obtained at feed OA concentration of 1 mM and HRT=1 h and feed OA concentration of 2 mM and HRT=2 h, respectively). However, the impact of O_3 dosage is less prominent at higher catalyst dosages with higher OA removal observed at higher catalyst loading, particularly for lower (< 7.5) feed O_3 :DOC concentration ratios. Note that only slight enhancement in OA removal by pure ozonation with increase in $O_3(g)$ dosage (by either varying the $O_3(g)$ concentration and/or $O_3(g)$ flow rate) was observed (Fig. S4), at least at the conditions investigated here, supporting the conclusion that the influence of O_3 dosage observed during HCO is mainly due to enhancement of the catalyst- O_3 interaction. As reported in our earlier work (Yuan et al., 2021), ozone adsorbs to the Cu-Al LDHs surface forming surficial ozone species which rapidly react with the surface-associated Cu-oxalate complexes. At the same feed O_3 :DOC concentration ratio, increasing the HRT showed an improvement in OA removal in the presence of $1.0 g.L^{-1}$ catalyst (Fig. 4d) suggesting that OA removal is limited by the reaction kinetics, possibly by the rate of adsorption of OA and/or O_3 on the Cu-Al LDHs surface and/or the $O_3(g)$ interphase mass transfer rate. We explore the rate-limiting step and optimize the operating conditions via use of kinetic modelling in the following section.

We would also like to highlight that there is no deterioration in the performance of the Cu-Al LDHs catalyst on short-term repeat use with removal efficiency maintained at $> 90 \%$ over 10 consecutive cycles of reuse (Fig. S5).

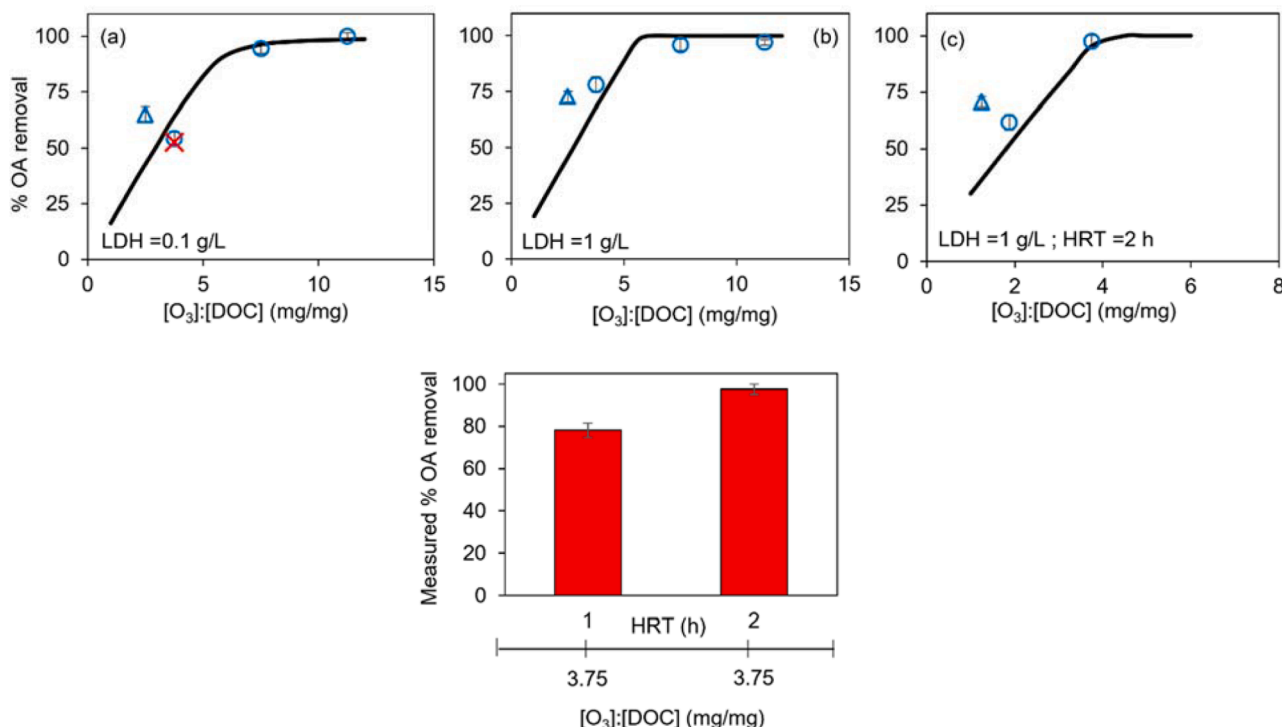


Fig. 4. Measured OA removal by HCO in the suspension reactor at varying $O_3:DOC$ ratio using 0.1 g/L (a) and 1.0 g/L (b & c) Cu-Al LDHs. $O_3:DOC$ concentration ratio was varied by either varying the $O_3(g)$ concentration (circles) or $O_3(g)$ flow rate (triangles) or OA concentration (cross). Panels b and c show the OA removal at 1 g/L Cu-Al LDHs at 1 mM OA and HRT = 1 h and 2 mM OA and HRT of 2 h, respectively. Panel d shows the effect of HRT on OA removal at $O_3:DOC$ concentration ratio of 3.75 for Cu-Al LDHs dosage = 1.0 g/L. Symbols represent average of measured values and error bars represent standard deviation of measurement. Lines represent model results as explained in detail in Section 3.5.

3.6. Development of kinetic model

A mathematical kinetic model was developed to explain the performance of the suspension reactor over a range of operating conditions by combining the model describing OA oxidation by Cu-Al LDHs (Table 1) developed in our earlier work (Yuan et al., 2021) with a simple mathematical expression of the ozone dissolution process (Eq. (3)). Note that for development of the kinetic model, we assumed that the suspension reactor is completely mixed with the particle suspension treated as a single (i.e. liquid) phase. As shown in Fig. 4a-c and 5a, the model describes the impact of various operating conditions on OA removal very well. The model also describes the outlet $O_3(g)$ concentration very well (Fig. 5b) with the agreement between measured and modelled results supporting the veracity of the ozone dissolution equation and

parameters used here. A good line of agreement ($R^2 > 0.96$) between the model-predicted value and experimental results shown in Fig. 5 suggests that the mathematical model developed here is robust and can be used to predict the performance of the suspension reactor. We would like to highlight that even though there is a small disparity (15–20%) between the model-predicted values and experimental data, particularly for low $O_3(g)$ flow rate conditions, the model effectively captures the overall trend of the experimental data presented here for a wide range of conditions used here. While the exact reason for the discrepancy in the model output and measured results, particularly for OA removal, is not clear, it could be related to the settling of some of the Cu-Al LDHs particles and/or slightly non-homogeneous mixing within the reactor at low $O_3(g)$ flow conditions.

Based on analysis of the rate of various reactions using the kinetic

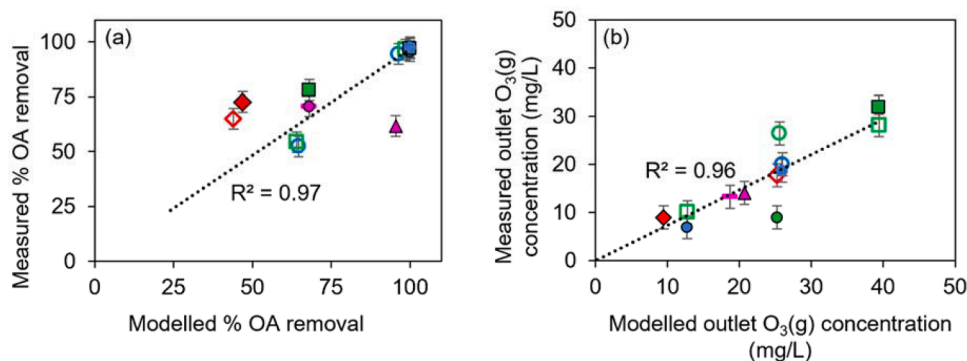


Fig. 5. Line of agreement between the model-predicted and measured OA removal % (a) and outlet $O_3(g)$ concentration (b) under varying operating conditions (circles: varying OA concentration, squares: varying $O_3(g)$ dosage, diamonds: varying $O_3(g)$ flow rate, triangles: varying HRT; open symbols and closed symbols represent results obtained at LDH dosage of 0.1 g/L and 1.0 g/L, respectively; pink symbols represent various $O_3(g)$ flow rate and $O_3(g)$ dosage conditions obtained at LDH dosage of 1.0 g/L, 2 mM OA and HRT of 2 h).

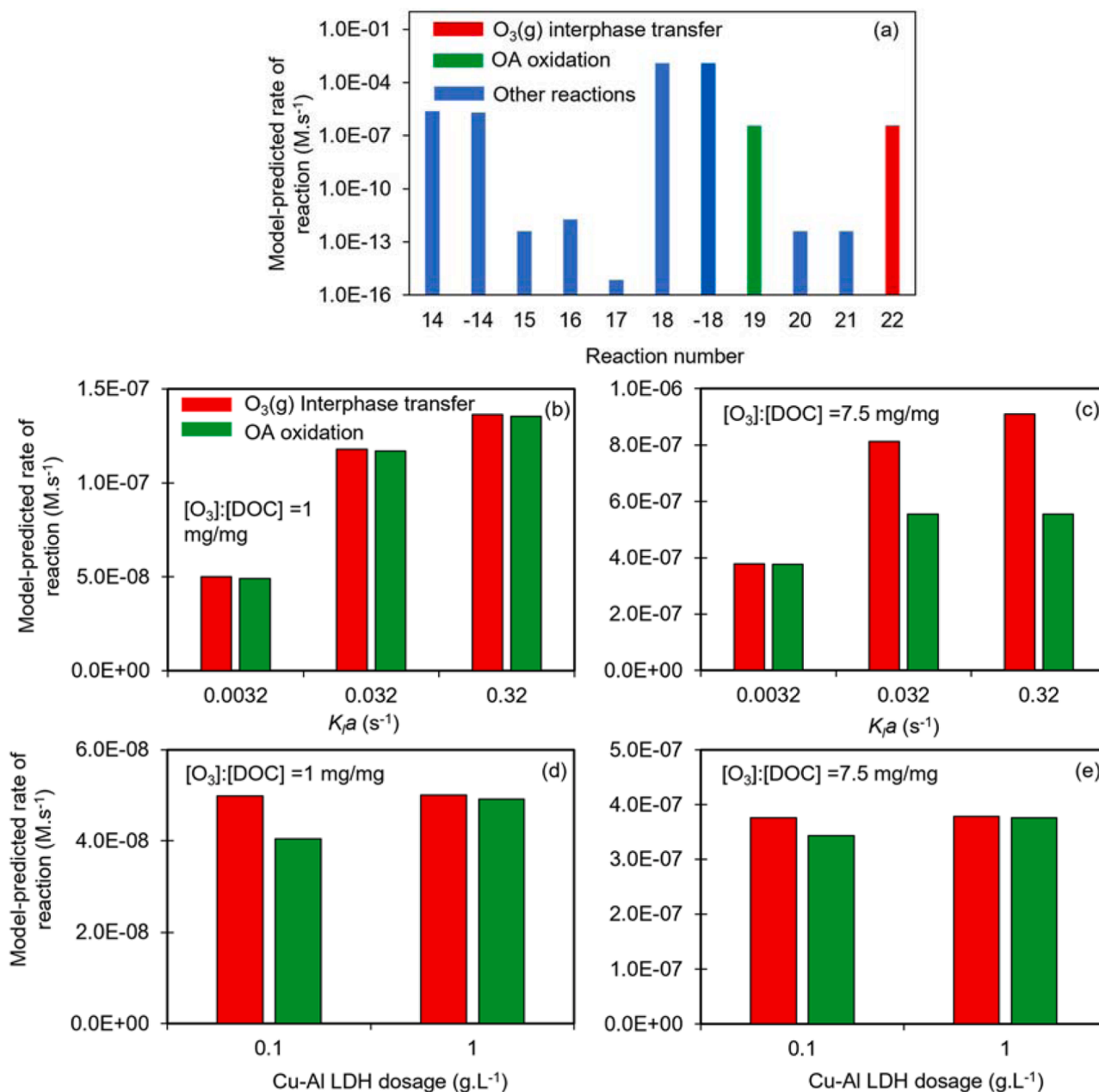


Fig. 6. Analysis of the rate of various reactions involved in OA oxidation by HCO using Cu-AL LDHs. Panel a shows the reaction rate of all the surface reactions highlighted in Table 1 (reactions 14–21) at $[O_3]:[DOC] = 7.5$ (feed $O_3(g) = 40$ mg/L, $O_3(g)$ flow rate = 3 L/min, HRT = 1 h, OA = 1 mM) and Cu-Al LDH dosage = 1.0 g/L at pH 8.5). Note that reaction 22 represents the ozone interphase transfer reaction. Panels b & c show the effect of varying the K_a value on the ozone interphase mass transfer rate (red bars) and the OA oxidation rate (green bars) at $[O_3]:[DOC]$ of 1 and 7.5, respectively. Panels d & e show the effect of varying the catalyst dosage on the ozone interphase mass transfer rate (red bars) and OA oxidation rate (green bars) at $[O_3]:[DOC]$ of 1.0 and 7.5 (mg/mg), respectively.

model developed here (Fig. 6), we further determine that the rate of OA removal in the present reactor is limited by the $O_3(g)$ interphase mass transfer rate. As indicated in Fig. 6b and 6c, increasing the interphase ozone mass transfer coefficient (i.e. K_a) by 10–100 times increases the $O_3(g)$ interphase mass transfer rate and (concomitantly) OA oxidation rate. The influence of K_a value on OA removal is more apparent at lower ozone dosage since at higher O_3 dosage, even though ozone dissolution is slow, complete removal of OA can still be observed due to the excess dosage of ozone. Increasing the catalyst dosage has no impact on the overall $O_3(g)$ interphase mass transfer rate and OA removal rate suggesting that surface reaction kinetics do not limit OA oxidation. As indicated in Fig. 6a, the rate of O_3 adsorption and concomitant decay as well as the rate of OA adsorption on the catalyst surface are much faster than the $O_3(g)$ interphase mass transfer rate further confirming that O_3

availability at the catalyst surface is limited by ozone dissolution. These insights suggest that improvement in ozone gas diffuser design is required to improve the performance of the suspension reactor.

Employing the kinetic model developed here, we have further predicted the influence of various operating conditions (catalyst dosage, O_3 :DOC concentration ratio (mg/mg) and HRT) on the reactor performance when treating feed waters containing an OA concentration of 1 mM (i.e., DOC = 24 mg/L) at pH 8.5. We use three parameters, namely, OA removal efficiency, ozone dissolution efficiency ($1 - [O_3(g)]_{out}/[O_3(g)]_{in}$) and ozone utilization efficiency (i.e., mg O_3 consumed per mg of DOC), to gauge the performance of the suspension reactor. As shown in Fig. 7, the model shows that as the O_3 :DOC concentration ratio increases, OA removal increases with OA removal $\geq 90\%$ achieved at O_3 :DOC concentration ratios ≥ 4.0 (mg/mg). Increasing the catalyst dosage has no

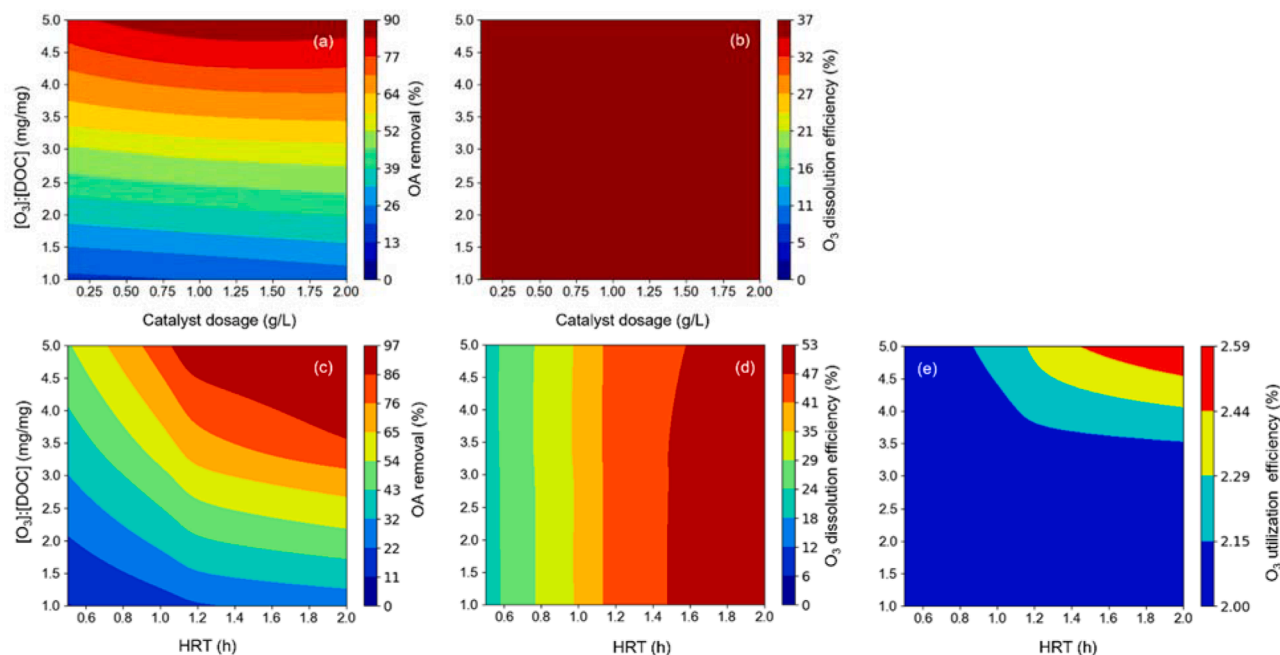


Fig. 7. Model-predicted OA removal efficiency (a) and O_3 dissolution efficiency (b) at varying catalyst dosage and $[O_3]:[DOC]$ at HRT = 1 h. Panels c-e show the OA removal efficiency (c) and O_3 dissolution efficiency (d) and O_3 utilization efficiency (e) at varying $[O_3]:[DOC]$ and HRT.

impact on the OA removal or ozone dissolution efficiency. In comparison, increase in HRT significantly impacts the OA removal and ozone dissolution efficiency. These results further support the conclusion that the kinetics of ozone dissolution limit the rate and extent of OA removal in the system. Increasing the reaction time facilitates dissolution of ozone and, concomitantly, OA removal. Under all conditions investigated, the ozone utilization efficiency was ~ 2 (mg/mg) which is close to stoichiometric constraints (i.e., 1 mole of O_3 required for 1 mole of OA removal) confirming that there is no futile consumption of dissolved ozone in the reactor system investigated here.

Overall, the developed kinetic modelling approach is useful for optimizing the operating conditions and predicting the performance of the reactor. Furthermore, the model developed here can be combined with reactor hydrodynamics using computational fluid dynamics modelling tools and used to optimize the design and performance of full-scale systems.

4. Conclusions

In this study, we compare the performance of fixed-bed and suspension HCO reactors with respect to ozone dissolution and organic removal efficiency, employing oxalic acid as the target organic compound. We used Fe-oxide@ Al_2O_3 and Fe-oxide@ Al_2O_3 /Cu-Al LDHs as the catalysts in fixed-bed and suspension reactors, respectively. Our results show that the suspension reactor displays a much higher ozone dissolution efficiency and concomitant OA removal compared to that observed in the fixed-bed reactor. The lower $O_3(g)$ dissolution efficiency in the fixed-bed reactor is possibly due to the large bubble size in the catalytic bed due to bubble retention with resultant low $O_3(g)$ interphase mass transfer rate. Ineffective utilization of the catalyst-bed further lowers O_3 -catalyst interaction and subsequent generation of oxidants and oxidation of organics. Given the promising performance of the HCO in suspension reactor using Cu-Al LDHs as the catalyst, we evaluated the influence of various operating conditions (ozone dosage, catalyst dosage, HRT, etc.) on the OA removal efficiency in the suspension reactor. Our results show that the $O_3:DOC$ concentration ratio influences the OA removal efficiency with increase in ozone dosage facilitating OA removal. The HRT also influences OA removal,

particularly at low ozone gas dosage; however, the influence of catalyst dosage (at least in the range investigated here) on OA removal is minimal. A reaction-kinetics based mathematical model was developed to predict the performance of the suspension reactor. Quantitative analysis of the reaction kinetics using the kinetic model developed here suggests that $O_3(g)$ interphase transfer is the rate-limiting step in OA removal. Hence, some modification to ozone diffuser design is required to further improve the performance of the suspension reactor.

Overall, this study shows that the suspension catalytic ozonation reactor configuration is more effective than the fixed-bed reactor configuration. As such, future studies should focus on the development of catalysts that can be used in a suspension reactor. While the Cu-Al LDHs catalyst developed here can be easily maintained in suspension and shows good OA removal efficiency, the performance of the catalyst for the treatment of real wastewaters that typically contain a mixture of organic compounds needs to be validated. While the pilot-scale system used here is well mixed with homogeneous distribution of catalyst and solution phase species, this is unlikely to be the case in full scale systems. Hence, coupling of the reaction kinetics model developed here with computational fluid dynamics modelling is needed to optimize the design and performance of industrial scale suspension reactors.

CRedit authorship contribution statement

Zarak Mahmood: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft. **Shikha Garg:** Conceptualization, Data curation, Formal analysis, Methodology, Writing – review & editing. **Yuting Yuan:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – review & editing. **Ling Xie:** Formal analysis, Investigation, Methodology. **Yuan Wang:** Conceptualization, Methodology, Project administration, Supervision, Writing – review & editing. **T David Waite:** Conceptualization, Methodology, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.watres.2024.121410](https://doi.org/10.1016/j.watres.2024.121410).

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